# Kennesaw State University 2026-2027 Undergraduate Catalog

## Courses

Contract All Courses |

## Mathematics

- ### MATH 0997: Support for Quantitative Reasoning

  **1 Credit Hours**

  *Corequisite:* MATH 1001

  **Description**
  This Learning Support course provides corequisite support in mathematics for students enrolled in MATH 1001 - Quantitative Reasoning. Topics will parallel topics being studied in MATH 1001 and the course will provide support for the essential quantitative skills needed to be successful in MATH 1001. Taken with MATH 1001, topics to be covered will include logic, basic probability, data analysis and modeling from data.

- ### MATH 0999: Support for College Algebra (MATH 1111)

  **1 Credit Hours**

  *Corequisite:* MATH 1111

  **Description**
  This course provides corequisite skills and additional instruction for topics and concepts covered in MATH 1111 College Algebra.

- ### MATH 1001: Quantitative Reasoning

  **3 Credit Hours**

**Description**
This course emphasizes quantitative reasoning skills needed for informed citizens to understand the world around them. Topics include logic, basic probability, data analysis and modeling from data.

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**Course Learning Outcomes**
Students who successfully complete this course will be able to:

1. Express information, concepts, and thoughts in verbal, numeric, graphical, and symbolic form while solving a variety of problems.
2. Manipulate information, concepts, and thoughts in verbal, numeric, graphical, and symbolic form while solving a variety of problems in real life context.
3. Solve problems through inductive and deductive reasoning; apply inductive and deductive reasoning to real life situations.
4. Model situations from a variety of contexts by extracting quantitative data, translating the data into information in various modes, evaluating the information, abstracting essential information, making logical deductions, and arriving at reasonable conclusions.
5. Use appropriate technology in the evaluation, analysis, and synthesis of information in problem solving situations.

- ## MATH 1101: Introduction to Mathematical Modeling

**3 Credit Hours**

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**Description**
This course is an introduction to mathematical modeling using graphical, numerical, symbolic, and verbal techniques to describe and explore real-world data and phenomena. Emphasis is on the use of elementary functions to investigate and analyze applied problems and questions, supported by the use of appropriate technology, and on effective communication of quantitative concepts and results.

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**Notes:** This course is managed through the cooperative academic agreement known as eCore.

- ## MATH 1111: College Algebra

**3 Credit Hours**

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**Description**
This course provides an in-depth study of the properties of algebraic, exponential and logarithmic functions as needed for calculus. Emphasis is on using algebraic and graphical techniques for solving problems involving linear, quadratic, piece-wise defined, rational, polynomial, exponential, and logarithmic functions.

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**Course Learning Outcomes**
Students who successfully complete this course will be able to:

1. Model situations from a variety of settings by using polynomial, exponential and logarithmic functions.

2. Manipulate mathematical information, concepts, and thoughts in verbal, numeric, graphical and symbolic form while solving a variety of problems which involve polynomial, exponential or logarithmic functions.
3. Apply a variety of problem-solving strategies, including verbal, algebraic, numerical, and graphical techniques, to solve multiple-step problems involving polynomial, exponential, logarithmic equations and inequalities and systems of linear equations.
4. Shift among the verbal, numeric, graphical and symbolic modes in order to analyze functions.
5. Derive common formulas, such as the quadratic formula and logarithmic identities.

- ## MATH 1113: Precalculus

**3 Credit Hours**

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*Prerequisite:* MATH 1111 or By Placement

**Description**

This course is an intensive study of the basic functions needed for the study of calculus. Topics include algebraic, functional, and graphical techniques for solving problems with algebraic, exponential, logarithmic, and trigonometric functions and their inverses.

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**Course Learning Outcomes**

Students who successfully complete this course will be able to:

1. Evaluate, analyze, manipulate, and create graphical representations of polynomial, rational, algebraic, exponential, logarithmic, trigonometric, and piecewise defined functions.
2. Use polynomial, rational, exponential, logarithmic, trigonometric, and piecewise defined functions as models for phenomena.
3. Use algebra of functions, function properties, and identities to solve applied problems.
4. Manipulate mathematical information in verbal, numerical, graphical, and symbolic form when solving applied problems.
5. Write mathematical expressions, equations, and problem solutions using proper, standard mathematical notation.

- ## MATH 1160: Elementary Applied Calculus

**3 Credit Hours**

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*Prerequisite:* MATH 1111, MATH 1113, or By Placement

**Description**

Uses techniques of college algebra and elementary calculus to analyze and model real world phenomena. The emphasis will be on applications using an intuitive approach to the mathematics rather than formal development. Topics include graphs, derivatives, and integrals of functions. The course incorporates collaborative learning, oral and written reports, and technology.

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**Notes:** This course is not intended for majors within the College of Science and Mathematics or the Southern Polytechnic College of Engineering.

**Course Learning Outcomes**

Students who successfully complete this course will be able to:

1. Evaluate limits of functions and determine the continuity of functions.
2. Find derivatives of algebraic, logarithmic, and exponential functions.
3. Evaluate integrals of basic algebraic and exponential functions.
4. Solve applied problems using derivatives and integrals, such as graphing, optimization, elasticity of demand, exponential growth and decay, the area between curves, average value, accumulated present and future value, and consumers' and producers' surplus.

## MATH 1179: Calculus I for Life Sciences

**4 Credit Hours**

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*Prerequisite:* MATH 1113, or By Placement

**Description**

This is a first course in Calculus in which special emphasis will be given to using the tools of calculus to build and analyze mathematical models, especially those arising in the life sciences. This course introduces the central concepts of single variable calculus including limits, derivatives, and integrals of algebraic and transcendental functions with particular motivations from and application to the Biological Sciences.

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**Notes:** Students completing this course may not also receive credit for MATH 1190. This course is not appropriate for mathematics, chemistry/biochemistry, physics, computer science, or engineering technology majors or minors. Credit for MATH 1189 can be combined with credit for MATH 1179 to satisfy the prerequisite for MATH 2202.

**Course Learning Outcomes**

Students who successfully complete this course will be able to:

1. Evaluate problems in life sciences and demonstrate how derivatives or integrals can help find solutions.
2. Explain the fundamentals of and evaluate limits, derivatives, and integrals using tabular, graphical, and algebraic techniques and examples.
3. Use the calculus tools of limits, derivatives, and integrals to build and analyze mathematical models, including differential equations, of problems that arise in the life sciences.

## MATH 1189: Bridge to Calculus II

**1 Credit Hours**

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*Prerequisite:* MATH 1179

**Description**

This 1-hour course is for students who have completed Calculus for Life Sciences and will prepare them to be successful in Calculus II. A brief review of Calculus I is provided with special emphasis on select topics that are not covered in MATH 1179. These include Continuity, Limits involving infinity, Calculus of Inverse

Trigonometric functions, The Mean Value Theorem, and L'Hôpital's rule.

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- ## MATH 1190: Calculus I

**4 Credit Hours**

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*Prerequisite:* MATH 1113, or By Placement

**Description**

This course is the first in the calculus curriculum and introduces the central concepts of calculus. Topics include limits, continuity, derivatives of algebraic and transcendental functions of one variable, applications of these concepts and a brief introduction to the integral of a function.

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**Course Learning Outcomes**

Students who successfully complete this course will be able to:

1. Evaluate limits and continuity of functions by using algebraic manipulations and looking at graphs and tables.
2. Calculate derivatives of algebraic, trigonometric, exponential, and logarithmic functions.
3. Use derivatives to solve applied problems.
4. Calculate integrals of algebraic, trigonometric, exponential, and logarithmic functions.
5. Use integrals to solve applied problems.

- ## MATH 1401: Elementary Statistics

**3 Credit Hours**

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**Description**

This is a non-calculus based introduction to statistics. Topics include descriptive statistics, probability, distributions, hypothesis testing, inferences, correlation, and regression.

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**Notes:** This course is managed through the cooperative academic agreement known as eCore.

- ## MATH 1501: Calculus I

**4 Credit Hours**

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*Prerequisite:* MATH 1113

**Description**

This course is a four (4) credit hour course and includes material on functions, limits, continuity, the derivative, anti-differentiation, the definite integral, and techniques of integration.

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**Notes:** This course is managed through the cooperative academic agreement known as eCore.

- ## MATH 2008: Foundations of Numbers and Operations

**3 Credit Hours**

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*Prerequisite:* The completion of Mathematics and Applied Math IMPACT Core courses with a minimum grade of "C" or higher, and enrolled in one of the following majors: Elementary Education, Elementary Education (Interest), Early Childhood Birth to Kindergarten Traditional, or Early Childhood Birth to Kindergarten Traditional (Interest).

**Description**

This course is an introductory mathematics course for early childhood education majors. The course emphasizes the understanding and use of the major concepts of number and operations. As a general theme, strategies of problem solving are used and discussed in the context of various topics.

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- ## MATH 2202: Calculus II

**4 Credit Hours**

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*Prerequisite:* MATH 1190 or (MATH 1179 and MATH 1189)

**Description**

This course is the second in the calculus curriculum and consists of two parts. The first part is concerned with the techniques of integration and applications of the integral. The second part is concerned with infinite sequences and series.

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**Course Learning Outcomes**

Students who successfully complete this course will be able to:

1. Evaluate a wide variety of standard integrals.
2. Apply integrals to a variety of problems involving area, volume, curve length, work done by a force and other similar quantities.
3. Understand the concept of convergence of improper integrals and be able to evaluate some basic improper integrals;
4. Understand the concept of convergence of infinite series and be able to derive and use a variety of tests for convergence of a series;
5. Understand the concept of a power series, be familiar with the important properties of power series, and understand the concept of the Taylor series of common functions.

- ## MATH 2203: Calculus III

**4 Credit Hours**

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*Prerequisite:* A grade of "C" or better in MATH 2202

**Description**

This course is the third in the calculus curriculum and is concerned with functions defined on regions in two or three dimensional space and that have values in one, two, or three dimensional space. Topics include partial derivatives, vector fields, multiple integrals, and applications of these topics.

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- ## MATH 2306: Ordinary Differential Equations

**3 Credit Hours**

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*Prerequisite:* A grade of "C" or better in MATH 2202

**Description**

An introduction to the theory of ordinary differential equations (ODEs), methods of solving first and higher order linear differential equations and linear systems, some applications in the sciences and engineering, the Laplace transform and its application in solving differential equations and linear systems, and Euler's numerical algorithm.

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- ## MATH 2335: Numerical Methods for Engineers

**3 Credit Hours**

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*Prerequisite:* A grade of 'C' or higher in MATH 2202, and ((CSE 1321 and CSE 1321L), or ECET 3710 or ME 1311)

**Description**

This course is an introduction to numerical approximation techniques in the solution of problems encountered in engineering and science. Topics include Taylor polynomials, iterative methods for root finding, interpolation, numerical quadrature and differentiation. Error analysis, effective application, and limitations of methods are emphasized. Notes: Not intended for mathematics or mathematics education majors.

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**Notes:** Not intended for mathematics or mathematics education majors.

- ## MATH 2345: Discrete Mathematics

**3 Credit Hours**

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*Prerequisite:* MATH 1113 or MATH 1190 or MATH 1179

**Description**

An introduction to the fundamentals of discrete mathematics. Topics include sets, formal logic, methods of proof, counting relations, functions, graphs and trees, and finite state automata.

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**Notes:** Not intended for mathematics or mathematics education majors.

- ## MATH 2390: Introduction to Logic, Set Theory, and Proofs

**3 Credit Hours**

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*Prerequisite:* MATH 2202

**Description**

This course introduces to students the foundations of logic, set theory, and basic proof techniques. The course serves as a bridge from the procedural and computational understanding of mathematics to a broad understanding encompassing logical reasoning, generalization, abstraction, axiomatic approach, and formal proof.

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- ## MATH 3000: Software of Mathematics

**3 Credit Hours**

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*Prerequisite:* Grades of "C" or better in MATH 2202 and (CSE 1321 and CSE 1321L)

**Description**

This course is designed to introduce students to numerical/symbolic computation using specialized mathematical software packages. The professional software to be taught may be MATLAB, MAPLE, Scientific Notebook or their equivalent. At the end of the course, students will be able to solve complex mathematical problems with the use of software and to write and present scientific or mathematical work professionally.

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- ## MATH 3204: Calculus IV

**3 Credit Hours**

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*Prerequisite:* MATH 2203

**Description**

This course is the fourth in the calculus curriculum and is concerned with the change of variables for integrals on two and three dimensional regions, line integrals, surface integrals, Green's theorem, and Stokes theorem. The analogue of Stokes' theorem (the theorem of Gauss) for integrals of functions on three-dimensional

parametric regions will also be studied.

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- ## MATH 3260: Linear Algebra I

**3 Credit Hours**

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*Prerequisite:* MATH 1190 or (MATH 1179 and MATH 1189)

**Description**

An introduction to linear algebra and some of its classical and modern applications. Among topics to be included will be systems of linear equations, matrices, determinants of matrices and applications, vector spaces, and inner product spaces. Significant use of technology will be employed in performing matrix computations.

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- ## MATH 3261: Numerical Methods

**3 Credit Hours**

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*Prerequisite:* MATH 3260 and (CSE 1321 and CSE 1321L)

**Description**

This course introduces fundamental numerical methods for solving nonlinear equations, systems of linear equations, and eigenvalue-eigenvector problems. Interpolation and approximation of functions will be considered. Computing and programming techniques will be used extensively.

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**Notes:** Extensive computer use will be incorporated

- ## MATH 3262: Mathematical Modeling

**3 Credit Hours**

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*Prerequisite:* MATH 2306

**Description**

This course is designed to introduce students to fundamental concepts and methods of mathematical modeling, through a project-oriented approach. This course will involve applications of mathematical techniques to solve problems in areas such as ecology, biology, finance, social sciences, life sciences, physical sciences and engineering. The emphasis will be on the building of mathematical models and on interpreting the solutions of these models in terms of real-life applications. The course will emphasize skills in constructing and analyzing models.

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- ## MATH 3272: Introduction to Linear Programming

**3 Credit Hours**

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*Prerequisite:* MATH 3260
**Description**
The simplex method, dual simplex method, dual-primal two phase method, and several interior-point methods for linear programming problems will be introduced.

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**Notes:** Selected applications will be discussed.

- ## MATH 3322: Graph Theory

**3 Credit Hours**

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*Prerequisite:* MATH 2345 or MATH 2390
**Description**
This course serves as an introduction to the basic principles of graph theory. Topics include but are not limited to graph representations, isomorphisms, paths, cycles, colorings, trees, matchings, planarity, graph algorithms, and optimization.

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- ## MATH 3324: Enumerative Combinatorics

**3 Credit Hours**

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*Prerequisite:* MATH 2345 or MATH 2390
**Description**
This course is an introduction to classical combinatorics and the theory of counting. Topics include the twelvefold way, combinatorial proof, the principle of inclusion/exclusion, and generating functions.

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- ## MATH 3332: Probability Theory

**3 Credit Hours**

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*Prerequisite:* MATH 2203
**Description**

This course is an introduction to probability theory. Topics include counting techniques, discrete and continuous univariate and multivariate random variables, expectation, moment generating functions, and the Central Limit Theorem.

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**Notes:** The MINITAB statistical software package is used.

- ## MATH 3396: Cooperative Study

**1-3 Credit Hours**

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*Prerequisite:* Approval of the coordinator of cooperative education/internship.

**Description**

A supervised work experience program for a minimum of two academic semesters at a site in business, industry or government. For sophomore, junior or senior level students who wish to obtain successive on the job experience in conjunction with their academic training.

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- ## MATH 3398: Internship

**1-9 Credit Hours**

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*Prerequisite:* Approval of major area committee and department chair.

**Description**

This course is a supervised, credit-earning work experience of one academic semester with a previously approved business firm, private agency, or government agency.

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- ## MATH 3405: Probabilistic Foundations of Actuarial Science

**3 Credit Hours**

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*Prerequisite:* MATH 2203 and MATH 3332

**Description**

This course serves as a preparation for Exam P of the Society of Actuaries. Emphasis is on joint continuous distributions, moment generating function, transformations and probability tools to assess risk.

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- ## MATH 3496: Elementary Number Theory

**3 Credit Hours**

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*Prerequisite:* MATH 2390

**Description**

The course is an introduction to the basic principles of number theory. Topics include properties of integers, congruences, divisibility, the Euclidean algorithm, prime number theorems, multiplicative functions, Diophantine equations, and applications in cryptology.

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- ## MATH 3696: College Geometry

**3 Credit Hours**

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*Prerequisite:* A grade of C or better in MATH 2202

**Description**

This is a rigorous development of geometry that starts with a close reading of Book I of Euclid's Elements, moves on to geometry developed during the Arabic period and the Renaissance, then to non-euclidean geometries discovered during the 19th century. The course includes a treatment of Hilbert's approach to Euclidean geometry and a brief treatment of real projective geometry. Students taking this course should have a serious interest in abstract mathematics.

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- ## MATH 4260: Linear Algebra II

**3 Credit Hours**

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*Prerequisite:* MATH 3260

**Description**

Topics in this course include real vector spaces and their subspaces; inner product spaces, orthogonal subspaces, Gram - Schmidt process; best approximation; eigenvalues and eigenvectors; special matrices; matrices of general transformations, and various applications including matrix functions.

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- ## MATH 4310: Partial Differential Equations

**3 Credit Hours**

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*Prerequisite:* MATH 2203 and MATH 2306

**Description**

This course is an introduction to partial differential equations (PDEs), their applications in the sciences and the techniques that have proved useful in analyzing them. The techniques include separation of variables, Fourier series and Fourier transforms, orthogonal functions and eigenfunction expansions, Bessel functions, and Legendre polynomials. The student will see how the sciences motivate the formulation of partial differential equations as well as the formulation of boundary

conditions and initial conditions. Parabolic, hyperbolic, and elliptic PDEs will be studied.

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- ## MATH 4345: Numerical Methods for Differential Equations

**3 Credit Hours**

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*Prerequisite:* MATH 2306 and (CSE 1321 and CSE 1321L)

**Description**

This course introduces methods for numerically solving initial value problems and boundary value problems that arise in ordinary and partial differential equations. Numerical differentiation and integration will be considered. Computing and programming techniques will be used extensively.

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**Notes:** Extensive use of computing will be incorporated.

- ## MATH 4361: Modern Algebra I

**3 Credit Hours**

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*Prerequisite:* MATH 2390 and MATH 3260

**Description**

An introduction to the fundamental structures of abstract algebra (groups, rings, and fields), the connections of these structures with the algebra studied at the elementary level, and the historical development of modern algebra. The emphasis in this course is on groups.

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- ## MATH 4362: Modern Algebra II

**3 Credit Hours**

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*Prerequisite:* MATH 4361

**Description**

A continuation of Modern Algebra I with an emphasis on rings and fields.

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- ## MATH 4381: Real Analysis I

**3 Credit Hours**

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*Prerequisite:* MATH 2390

**Description**

This course provides a rigorous introduction to the calculus of a single real variable and a deeper awareness of the theory of calculus than can be achieved in the elementary calculus courses. Among the topics covered in the course are completeness of the number system, elementary topology of the real line, limits of sequences, and limits and continuity of functions. The aim of this course is two-fold, to provide an understanding of the nature of the real number system and its role in the theory of calculus, and to provide a training in the discovery and writing of rigorous mathematical proofs.

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- ## MATH 4382: Real Analysis II

**3 Credit Hours**

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*Prerequisite:* MATH 4381

**Description**

This course is a continuation of the study of functions of a real variable (Real Analysis I). Topics include the Riemann/Darboux integral, differentiability, sequences and series of functions. The aim of the course is to provide the students with a deeper understanding of the notions of sequences/series, integrability, and differentiability of functions of a real variable, as well as their properties and interconnections. While developing these concepts, we will focus on understanding and writing formal proofs, as well as emphasize their applications.

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- ## MATH 4391: Complex Analysis

**3 Credit Hours**

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*Prerequisite:* MATH 2203

**Description**

This course is an introduction to the basic concepts of complex analysis, its beautiful theory and powerful applications. Topics covered will include: the algebra and geometry of the complex plane, properties of elementary functions of a complex variable, analytic and harmonic functions, conformal mappings, continuity, differentiation, integration (Cauchy integral theory), singularities, Taylor and Laurent series, residues and, time permitting, their applications.

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- ## MATH 4400: Directed Study

**1-3 Credit Hours**

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*Prerequisite:* Approval of the instructor, major area committee, and department chair.
**Description**
Special advanced topics external to regular course offerings.

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- ## MATH 4490: Special Topics in Mathematics

**1-6 Credit Hours**

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*Prerequisite:* Approval of the instructor and department chair.
**Description**
This course is comprised of special selected topics of interest to faculty and students.

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- ## MATH 4491: Advanced Topics in Mathematics

**3 Credit Hours**

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*Prerequisite:* Permission of Department Chair
**Description**
This course is offered occasionally on topics proposed by the instructor. Students will be able to apply for admission to the course and the department chair will evaluate the application based on the sufficient measures and criteria for admission that have been determined by the instructor.

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- ## MATH 4596: Topology

**3 Credit Hours**

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*Prerequisite:* MATH 2390
**Description**
This course is an introduction to the study of topology. Topics include topological spaces, subspaces, basis, continuity, separation and countability axioms, connectedness, and compactness.

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- ## MATH 4699: Undergraduate Research

**3 Credit Hours**

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*Prerequisite:* Approval of the instructor.

**Description**

The student conducts original research in mathematics under the supervision of a faculty member. This research is the result of sustained effort on a problem in mathematics, either theoretical or applied. Making an original contribution to the field of mathematics is the focus of the course rather than a survey of existing work. Results are disseminated by on campus presentations, conference presentations and/or peer-reviewed journal publications.

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Contract All Courses |